

Actividad de aprendizaje 06

En mongodb Atlas, cargue las base de datos de ejemplo

The screenshot shows the 'Database Deployments' page in MongoDB Atlas. The left sidebar contains navigation links for 'DEPLOYMENT', 'Database', 'Data Lake', 'SERVICES', 'Triggers', 'Data API', 'Data Federation', 'Search', 'SECURITY', 'Quickstart', 'Backup', 'Database Access', 'Network Access', and 'Advanced'. The main content area displays 'Cluster0' with buttons for 'Connect', 'View Monitoring', 'Browse Collections', and a menu icon (annotated with a red circle 1). The menu is open, showing options: 'Edit Configuration', 'Command Line Tools', 'Load Sample Dataset' (annotated with a red circle 2), and 'Terminate'. Below the menu, there are performance metrics and a table of deployment details.

VERSION	REGION	CLUSTER TIER	TYPE	BACKUPS	LINKED APP SERVICES	ATLAS SQL	ATLAS SEP
6.0.5	AWS / N. Virginia (us-east-1)	M0 Sandbox (General)	Replica Set - 3 nodes	Inactive	Application-0	Connect	Create I

The screenshot shows the 'Load Sample Dataset' dialog box. It contains the text: 'We've created a sample dataset to help you test features on Cluster0.' Below this, it says 'Sample Dataset' and 'Size: ~350 MB'. At the bottom, there are two buttons: 'Cancel' and 'Load Sample Dataset' (annotated with a red circle 3).

Verifique que tenga la bd **sample_training** y la colección **companies**

The screenshot shows the 'Collections' page in MongoDB Atlas. The left sidebar shows a list of databases and collections, with 'sample_training' and 'companies' highlighted. The main content area displays the 'sample_training.companies' collection. It shows metadata: 'STORAGE SIZE: 18.03MB', 'LOGICAL DATA SIZE: 34.79MB', 'TOTAL DOCUMENTS: 9500', and 'INDEXES TOTAL SIZE: 396KB'. Below this, there are tabs for 'Find', 'Indexes', 'Schema Anti-Patterns', 'Aggregation', 'Search Indexes', and 'Charts'. A search bar is present with the text 'Type a query: { field: 'value' }'. Below the search bar, a sample document is displayed:

```
{
  "twitter_username": "BachelorWetpaint",
  "category_code": "web",
  "number_of_employees": 47,
  "founded_year": 2005,
  "founded_month": 10,
  "founded_day": 17,
  "deadpooled_year": 1,
  "tag_list": "wiki, seattle, elowitz, media-industry, media-platform, social-distrib...",
  "alias_list": "",
  "email_address": "info@wetpaint.com",
  "phone_number": "206.859.6300",
  "description": "Technology Platform Company"
}
```

At the bottom, it says '1-20 of many results'.

Usando MongoSH, realice las siguientes operaciones:

Conectarse a su cluster de MongoDB Atlas

```
Current Mongosh Log ID: 645b0a68739dd2fd036f0d3e
Connecting to:      mongodb+srv://<credentials>@cluster0.fdkccg1.mongodb.net/?appName=mongosh+1.8.0
Using MongoDB:      6.0.5 (API Version 1)
Using Mongosh:      1.8.0

For mongosh info see: https://docs.mongodb.com/mongosh-shell/

Atlas atlas-sx70sx-shard-0 [primary] test> show dbs
prueba                196.00 KiB
sample_airbnb          52.88 MiB
sample_analytics       9.13 MiB
sample_geospatial     1.38 MiB
sample_guides          40.00 KiB
sample_mflix           52.80 MiB
sample_restaurants     7.73 MiB
sample_supplies        1.14 MiB
sample_training        57.18 MiB
sample_weatherdata     2.79 MiB
admin                 372.00 KiB
local                  5.39 GiB
```

Activar la base de datos sample_training

```
Atlas atlas-sx70sx-shard-0 [primary] test> use sample_training
switched to db sample_training
Atlas atlas-sx70sx-shard-0 [primary] sample_training> show collections
companies
grades
inspections
posts
routes
trips
zips
Atlas atlas-sx70sx-shard-0 [primary] sample_training> _
```

Ingresa 2 nuevas compañías

```
db.companies.insertOne({"name":"MiempresaUPN",
"homepage_url":"http://miempresaupn.com",
"blog_url":"http://digitalupn.net/",
"twitter_username":"MiempresaUPN",
"category_code":"web",
"number_of_employees":20,
"founded_year":2012,
"email_address":"info@miempresaupn.com",
"description":"Technology Platform Company"})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.insertOne({"name":"MiempresaUPN",
... "homepage_url":"http://miempresaupn.com",
... "blog_url":"http://digitalupn.net/",
... "twitter_username":"MiempresaUPN",
... "category_code":"web",
... "number_of_employees":20,
... "founded_year":2012,
... "email_address":"info@miempresaupn.com",
... "description":"Technology Platform Company"})
{
  acknowledged: true,
  insertedId: ObjectId("645b0ce549259696068d33a6")
}
```

```
db.companies.insertOne({"name":"segundaEmresa",
"category_code":"web",
"number_of_employees":20,
"email_address":"correo@segunda.com",
"description":"Empresa matequilla"})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.insertOne({"name":"segundaEmresa",
... "category_code":"web",
... "number_of_employees":20,
... "email_address":"correo@segunda.com",
... "description":"Empresa matequilla"})
{
  acknowledged: true,
  insertedId: ObjectId("645b0d7949259696068d33a7")
}
```

Realice las siguientes consultas:

Muestre las compañías cuyo año de fundación(`founded_year`) sea mayor igual al 2013

```
db.companies.find({founded_year:{$gte:2013}})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.find({founded_year:{$gte:2013}})
[
  {
    _id: ObjectId("52cdef7c4bab8bd6752982d4"),
    name: 'Advaliant',
    permalink: 'advaliant',
    crunchbase_url: 'http://www.crunchbase.com/company/advaliant',
    homepage_url: 'http://www.advaliant.com',
    blog_url: 'http://blog.advaliant.com',
    blog_feed_url: 'http://blog.mediatrust.com/feed/',
    twitter_username: 'advaliant',
    category_code: 'advertising',
    number_of_employees: 60,
    founded_year: 2013,
    founded_month: 4,
    founded_day: 16,
    deadpooled_year: 2009,
    deadpooled_month: 5
  }
]
```

Si queremos contar cuantos son

```
db.companies.find({founded_year:{$gte:2013}}).count()
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.find({founded_year:{$gte:2013}}).count()
5
```

Muestre los campos `name`, `homepage_url`, `email_address`, `phone_number`, de las compañías cuyo numero de empleados sea menor a 20

```
db.companies.find({number_of_employees:{$lt:20}},{_id:0,name:1,hhomepage_url:1,
email_address:1,phone_number:1})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.find({number_of_employees:{$lt:20}},{_id:0,name:1, homepage_url:1, email_address:1, phone_number:1})
[
  {
    name: 'YouTube',
    homepage_url: 'http://www.youtube.com',
    email_address: '',
    phone_number: '(650) 253-0000'
  },
  {
    name: 'AdaptiveBlue',
    homepage_url: 'http://www.getglue.com',
    email_address: 'support@getglue.com',
    phone_number: ''
  },
  {
    name: 'GrandCentral',
    homepage_url: 'http://grandcentral.com',
    email_address: 'info@grandcentral.com',
    phone_number: '510 559 1212'
  },
  {
    name: 'Ikan',
    homepage_url: 'http://www.ikan.net',
    email_address: 'customerservice@ikan.net',
    phone_number: '887.369.4526'
  },
  {
    name: 'Pounce',
    homepage_url: 'http://pounce.com',
    email_address: 'info@pounce.com',
    phone_number: ''
  },
  {
    name: 'Mashery',
    homepage_url: 'http://www.mashery.com',
    email_address: 'support@mashery.com',
    phone_number: '(415) 684-7200'
  }
]
```

Existen 2930 documentos

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.find({number_of_employees:{$lt:20}},{_id:0,name:1, homepage_url:1, email_address:1, phone_number:1}).count()
2930
```

Agregue el campo eliminar con el valor de "SI" para las compañías con categoría WEB.

```
db.companies.updateMany({category_code:"web"},{$set:{eliminar:"SI"}})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.updateMany({category_code:"web"},{$set:{eliminar:"SI"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1949,
  modifiedCount: 1949,
  upsertedCount: 0
}
```

Elimine los documentos cuyo campo eliminar esté en SI

```
db.companies.deleteMany({eliminar:"SI"})
```

```
Atlas atlas-sx70sx-shard-0 [primary] sample_training> db.companies.deleteMany({eliminar:"SI"})
{ acknowledged: true, deletedCount: 1949 }
```